WindCtrl Validator

Controller Validation Report

Engineering demonstration

Scope and assumptions

Offline row-level validation of simulated or user-supplied test records. This is an independent portfolio project, not a turbine certification tool. Demonstration limits are configurable and are not manufacturer specifications. Controller output is interpreted as a power command in kW. Each record is checked for data validity, range, state consistency, temperature and tracking error.

Configuration

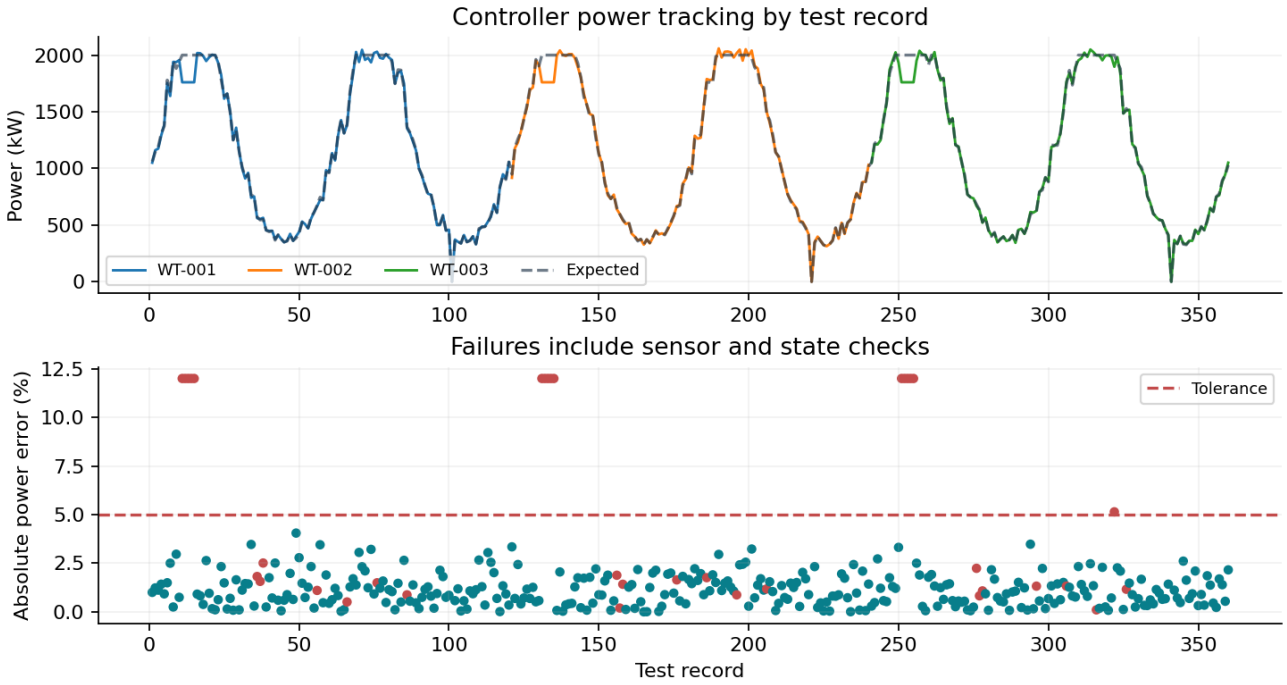
Power and controller tolerance: 5.0%. Zero-demand absolute tolerance: 0.5 kW. Maximum generator temperature: 100.0 degrees C. Boundaries are inclusive.

Validation summary

Metric	Value
Total Tests	360
Passed Tests	323
Failed Tests	37
Pass Pct	89.72
Average Error Pct	1.6019
Maximum Error Pct	12.0
Maximum Deviation Kw	240.0
Percentage Comparable Rows	357

Metric denominator. Percentage error excludes zero or invalid expected power and invalid observed power. Sensor failures remain included when power values are valid. Zero-demand records use absolute error; consult the comparable row count. Maximum deviation in kW includes valid zero-demand comparisons.

Engineering plots



Failure detail (first 20)

ID	Turbine	Reasons
11	WT-001	power deviation exceeds tolerance
12	WT-001	power deviation exceeds tolerance
13	WT-001	power deviation exceeds tolerance
14	WT-001	power deviation exceeds tolerance
15	WT-001	power deviation exceeds tolerance
36	WT-001	generator over-temperature
37	WT-001	generator over-temperature
38	WT-001	generator over-temperature
56	WT-001	wind_speed_mps: missing or non-finite number
66	WT-001	unexpected controller state; positive demand requires RUNNING state
76	WT-001	rotor_speed_rpm: outside demonstration range [0, 30]
86	WT-001	controller deviation exceeds tolerance
131	WT-002	power deviation exceeds tolerance
132	WT-002	power deviation exceeds tolerance
133	WT-002	power deviation exceeds tolerance
134	WT-002	power deviation exceeds tolerance
135	WT-002	power deviation exceeds tolerance
156	WT-002	generator over-temperature
157	WT-002	generator over-temperature
158	WT-002	generator over-temperature

Full results: `validation_results.csv`. Structured metrics: `summary.json`. The application does not model controller dynamics, transient response, interlocks or hardware-in-the-loop behaviour.